

MATH 140 Fall 2026

notes by

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A note on these notes. After each class, this document will be updated with the new material that was just covered. The timestamps in the left margin indicate when the notes from each day start. Subsections labelled with a * are optional. These notes should *not* be used as a substitute for the textbook. The best way to do well in this class is to do as many exercises from the textbook as you can stomach. This document may contain errors, typographical or mathematical. Please email me if you find any.

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1. Rates of change

This set of notes assumes that the reader is familiar with all the pre-calculus concepts that one learns in the later years of secondary school: functions, trigonometry, exponential functions, logarithms, etc. However, given that the very definition of what differential calculus is requires us to talk about functions and their inputs, it is perhaps prudent to begin with a refresher on real-valued functions with real-valued outputs.

A *function* is a collection of ordered pairs (x, y) with the rule that if (x, y) and (x, z) are both in the collection, then $y = z$. We can view x as the *input* to the function, and y as the *output* of the function; under this view, the rule above is simply that the function never outputs two different values when given the same input. People like to use the letter f to denote a function, and $f(x)$ to denote its value evaluated at a given input x . However, many particular functions are difficult to write without involving the input variable x in the expression, and in these cases we'll write $f(x)$ to mean the function as well.

Here are some examples of functions:

- The function $f(x) = 3x$, defined on all real numbers, that outputs triple its input value.
- The function that assigns to each real number x its square x^2 .

- c) The function $y = f(x)$ that equals 0 whenever x is a negative real number, and equals x whenever $x \geq 0$.
- d) The function that takes as input any nonzero real number x and outputs its reciprocal $1/x$.
- e) The function $f(x)$, defined on all real numbers, given by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational;} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Note that in each case above, we were careful to specify what the inputs to the function were. Going back to the definition of a function as a collection of ordered pairs (x, y) , the set of all x such that (x, y) is in the function is called the *domain* of f . (The set of all y such that (x, y) is in f is called the *range* of f , but this definition won't be as important to us.)

The title of this section is “rates of change”, and essentially this describes what this entire class is about. We want to study how much the output of a function changes when you change its input.

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